

IGSE

8. Indicadores energéticos y económicos comunitarios por tipo de usuario

Lecturer: Matteo Ranaboldo

INDEX

- Cálculos indicadores energéticos y económicos (globales)
- Cálculos indicadores energéticos y económicos por tipo de usuario
- Ejemplo de resultados

Input Data

Input data

- Hourly consumption of each user type (HC_i)
- Hourly wind turbine and solar generation (WG, SG) including losses
- Hourly energy buying (HBT) and selling tariffs (HST)

Variables

- Number of user of each type (NU_i)
- Nominal power of solar panels (NS)
- Number of wind turbines to be used (NW)

Energy Calculations

- Total hourly consumption in kWh (THC) $THC = \sum_{\forall i} (NU_i \times HC_i)$
 - Total hourly generation in kWh (THG) $THG = NW \times WG + NS \times SG$
 - Net hourly consumption in kWh (NHC) $NHC = THC - THG$
 - Overgeneration in kWh ($Surplus$) $Surplus = 0$ if $NHC > 0$
 $Surplus = abs(NHC)$ if $NHC < 0$
 - Undergeneration in kWh ($GridCons$) $GridCons = NHC$ if $NHC > 0$
 $GridCons = 0$ if $NHC < 0$
 - Self-Consumption in kWh ($SelfCons$) $SelfCons = \min(THC, THG)^*$
- $$TYC = \sum_{\forall h} THC$$
- $$TYG = \sum_{\forall h} THG$$
- $$YearSelfCons = \sum_{\forall h} SelfCons$$

Yearly Indicators [%]

* This is the maximum amount of energy the community can self-consume

- Self-Consumption (generation) (SCG) $SCG = \frac{YearSelfCons}{TYG}$
- Self-consumption (cons) ($autarky$) $Autarky = \frac{YearSelfCons}{TYC}$

Economic Calculations

- Hourly Buying price (HB) $HB = GridCons \times HBT$

- Hourly Selling price (HS) $HS = Surplus \times HST$

- Hourly Bill ($Hbill$) $HBill = HB - HS$

$$YBill = \sum_{\forall h} HBill$$

- Annual savings** $AnnualSaving = Ybill(noEC)^* - Ybill$

* $Ybill(noEC)$ is the yearly bill paid by all the users of the community before being part of the energy community, i.e. the sum of the initial bills of each user

- Installation cost** $InstalCost = SolarPlanelas + WindTurbines\ costs$

- Return period** $Payback = \frac{InstalCost}{AnnualSavings}$

- Levelized cost of energy (LCOE) to compare technologies, independently on the users connected

$$LCoE \& LCoH = \frac{Total\ lifetime\ cost}{Total\ lifetime\ energy\ production} = \frac{\sum_{t=1}^n \frac{I + FO\&M_t + VO\&M_t + F_t}{(1+r)^t}}{\sum_{t=1}^n \frac{E_t}{(1+r)^t}}$$

Where:

- I : Investment costs
- $FO\&M_t$: Fixed operation and maintenance costs in the year t
- $VO\&M_t$: Variable operation and maintenance costs excluding fuel costs in the year t
- F_t : Fuel costs in the year t
- E_t : Energy production in the year t
- r : Discount rate
- n : Expected asset lifetime

Energy Calculations per User

- Total yearly consumption of user type i

$$YC_i$$

- Yearly % of consumption of user type i

$$Yearly\%Cons_i = \frac{YC_i}{TYC}$$

- Hourly % of consumption of user type i

$$Hourly\%Cons_i = \frac{HC_i}{THC}$$

- Esta formula considera coeficientes de reparto dinamicos, o sea el % horario de cada usuarios no es fijo sino que cada hora cada usuario recibe proporcionalmente a su consumo (de esta forma se consume el maximo de la energia producida)
- Alternativamente se pueden considerar coeficientes estaticos, o sea cada usuario recibe un % fijo cada hora

- Hourly Generation distributed to user type i

$$HG_i = THG * Hourly\%Cons_i$$

$$YG_i = \sum_{\forall h} HG_i$$

- Net hourly consumption of user type i (NHC_i)

$$NHC_i = HC_i - HG_i$$

- Hourly Self-Consumption of user type i ($SelfCons_i$)

$$SelfCons_i = \min(HC_i, HG_i)$$

$$YearSelfCons_i = \sum_{\forall h} SelfCons_i$$

- Overgeneration in kWh ($Surplus_i$)

$$Surplus_i = 0 \text{ if } NHC_i > 0$$

$$Surplus_i = abs(NHC_i) \text{ if } NHC_i < 0$$

- Undergeneration in kWh ($GridCons_i$)

$$GridCons_i = NHC_i \text{ if } NHC_i > 0$$

$$GridCons_i = 0 \text{ if } NHC_i < 0$$

Yearly Indicators [%]

- Self-Consumption (generation) (SCG_i)

$$SCG_i = \frac{YearSelfCons_i}{YG_i}$$

- Self-consumption (cons) ($autarky_i$)

$$Autarky_i = \frac{YearSelfCons_i}{YC_i}$$

Economic Calculations per User

- Hourly Buying price of user type i (HB_i) $HB_i = GridCons_i \times HBT$
- Hourly Selling price of user type i (HS_i) $HS_i = Surplus_i \times HST$
- Hourly Bill ($Hbill_i$) $HBill_i = HB_i - HS_i$ $YBill_i = \sum_{\forall h} HBill_i$
- Annual savings** $AnnualSaving_i = Ybill(noEC)_i^* - Ybill_i$
- Installation cost** $InstalCost_i = \frac{InstalCost * Yearly\%Cons_i}{NU_i}$ NU_i is the number of users of type i
- Return period** $Payback_i = \frac{InstalCost_i}{AnnualSaving_i}$

* $Ybill(noEC)$ is the yearly bill paid by the user type i of the community before being part of the energy community, i.e. the sum of the initial bills of each user

Tarea a realizar

- 1) Desarrollar una herramienta para calcular indicadores energéticos y económicos por tipo de usuario
- 2) Evaluar los resultados considerando diferentes escenarios (ejemplos a continuación)
 - 1) Tipo de generación considerado (solo solar, solo eólica o ambas)
 - 2) Tipo de usuarios considerados (solo un tipo de usuario, diferentes tipos de usuarios)
 - 3) Número de usuarios de cada tipo
 - 4) Diferentes tarifas de electricidad
- 3) Discutir críticamente los resultados obtenidos

Energy community example

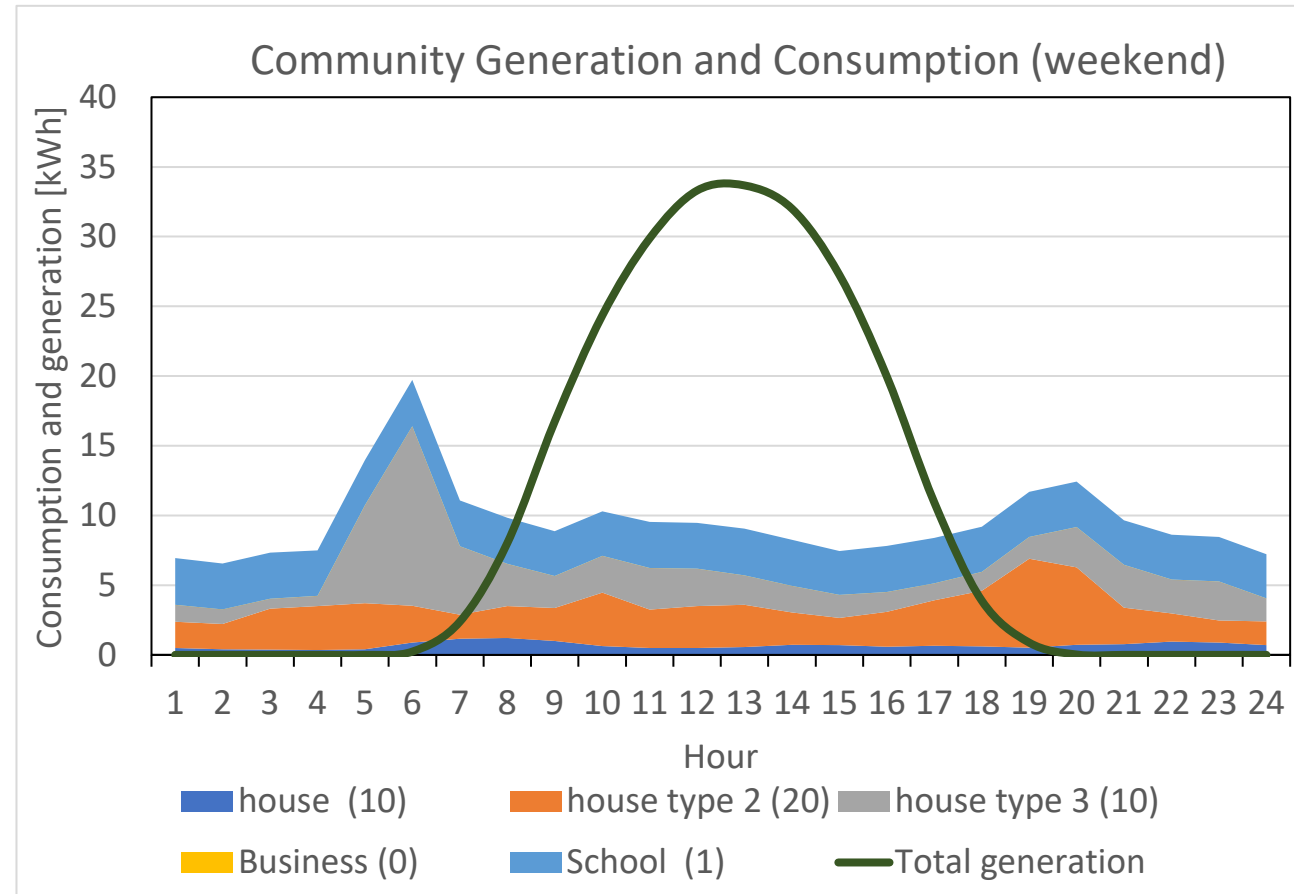
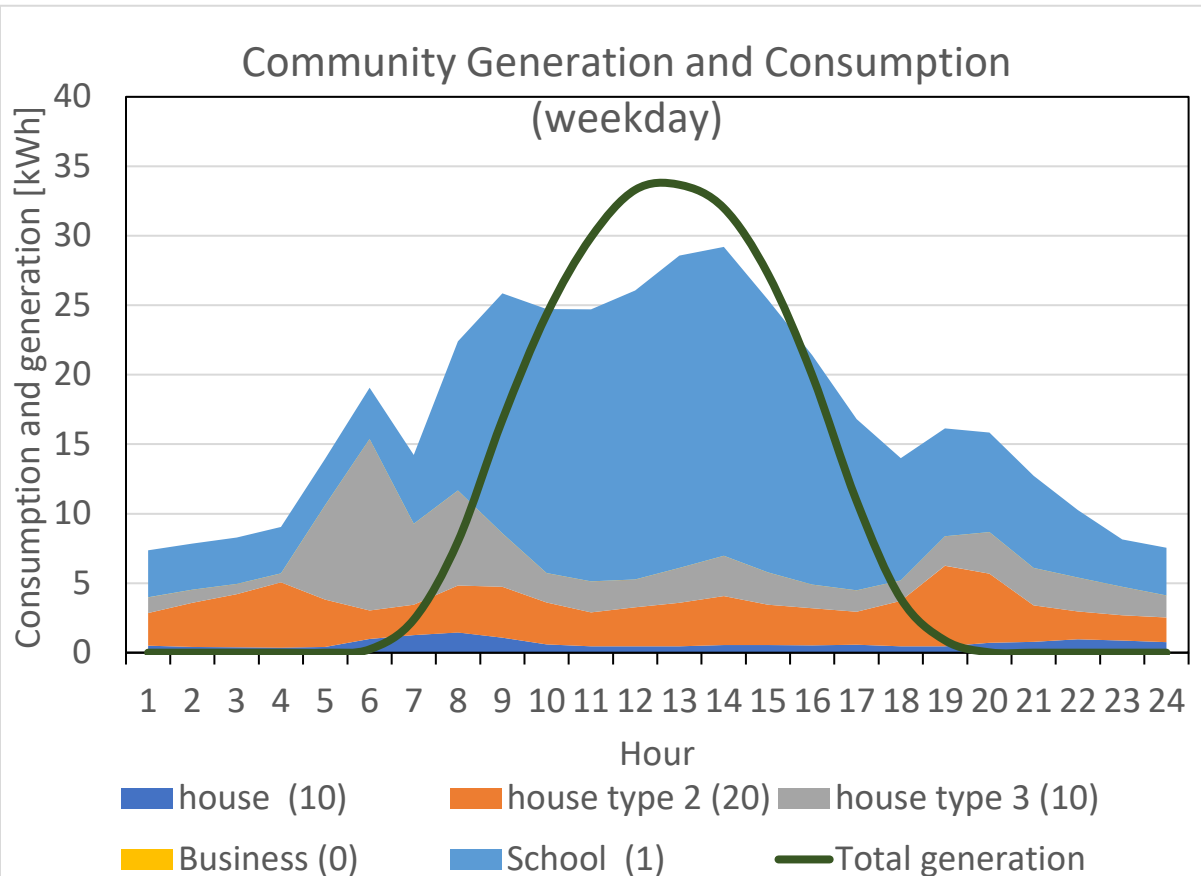
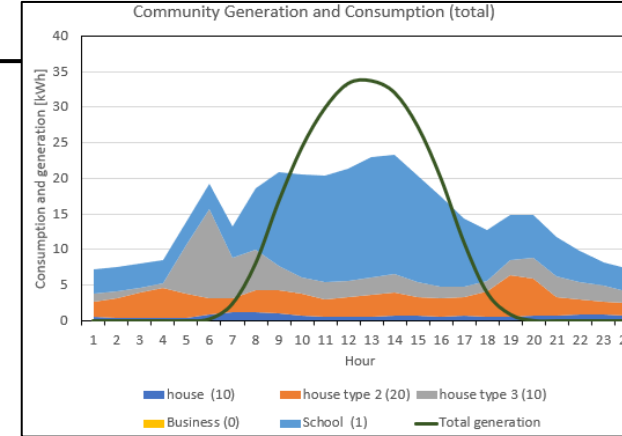
- Solar panels installed on a school and shared between different consumers
- Generation: 70 kW of installed PV power
- Consumption:
 - School
 - Association
 - Houses (25% low consumption, 50% average consumption, 25% high consumption)

Energy community example

			Tipus de comunitat		
		Consumption (kWh/y)	Scenario 1	Scenario 2	Scenario 3
Nombre de consumidors de cada tipus en cada escenari	Cases consum baix	594	10	20	20
	Cases consum mig	1320	20	40	40
	Cases consum alt	2565	10	20	20
	Associacions	3387	0	0	2
	Escola	72684	1	1	1
	Nombre total de consumidors a cada escenari		41	81	83
Total consumption			130.661 kWh/y	188.639 kWh/y	195.413 kWh/y
Solar generation (70 kW)			88.880 kWh/y		
Installation cost			77.000 €		
Self-consumption (% of generation)			61%	70%	72%
Self-consumption (% of consumption)			42%	33%	33%
Payback time			6.3 y	5.7 y	5.6 y

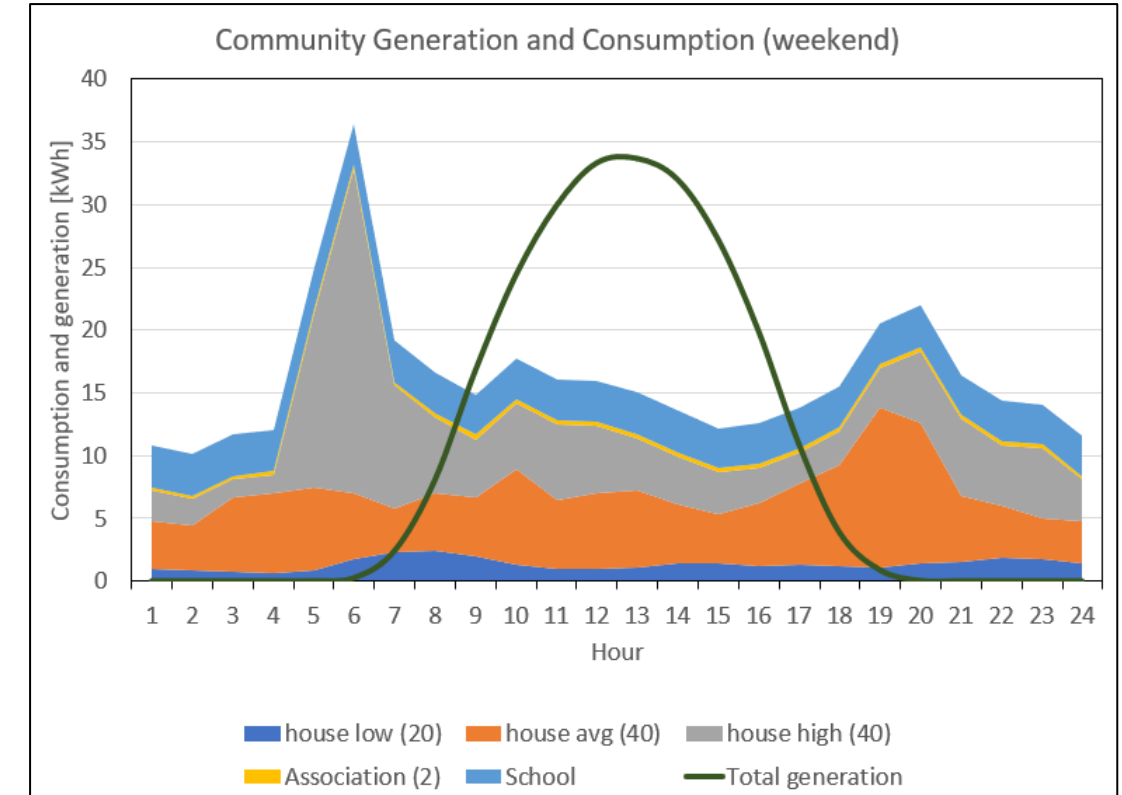
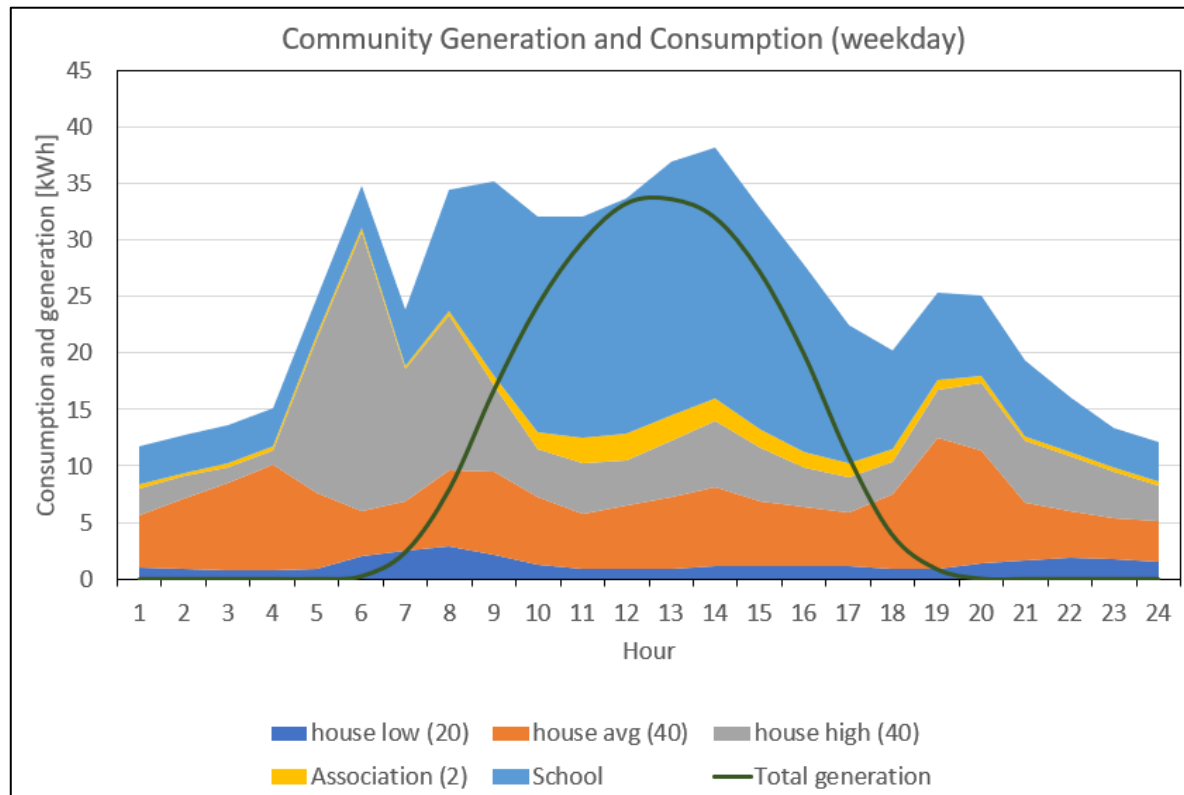
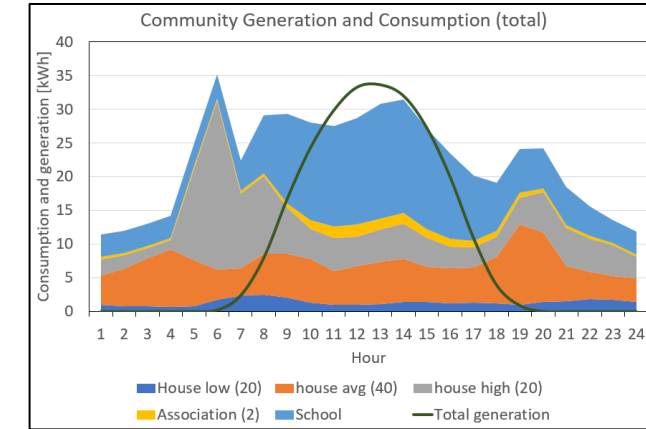
Energy community example

Scenario 1



Energy community example

Scenario 3



Energy community example

Scenario 3 results per user

	house low (20)	house avg (40)	house high (40)	Association (2)	School
total consumption [kWh/y]	11877.7	52780.0	51297.2	6774.0	72684.0
% consumption	6.1%	27.0%	26.3%	3.5%	37.2%
received energy [kWh]	5646.9	20834.6	15126.9	4738.5	42532.7
% received energy	6.4%	23.4%	17.0%	5.3%	47.9%
self-consumption [kWh]	3367.5	13931.6	11058.4	3141.8	32647.4
Self-consumption (gen) [%]	59.6%	66.9%	73.1%	66.3%	76.8%
Self-consumption (cons) [%]	28.4%	26.4%	21.6%	46.4%	44.9%
Buy per user [euros]	89.8	201.9	404.6	392.4	8390.4
Sell per user [euros]	4.5	6.9	7.8	34.4	419.8
Bill per user [euros]	207.6	317.3	519.1	517.7	8205.1
Bill - No Gen per user [euros]	243.0	390.7	632.7	893.5	15202.9
Savings per user [euros]	35.4	73.4	113.6	375.8	6997.8
Savings per user [%]	15%	19%	18%	42%	46%
initial inversion [euros]	234.0	519.9	1010.7	1334.6	28640.2
return period [years]	6.6	7.1	8.9	3.6	4.1